

Known Plaintext Attack on Hill Cipher

13. Write a C program for Hill cipher succumbs to a known plaintext attack if sufficient plaintext–ciphertext pairs are provided. It is even easier to solve the Hill cipher if a chosen plaintext attack can be mounted.

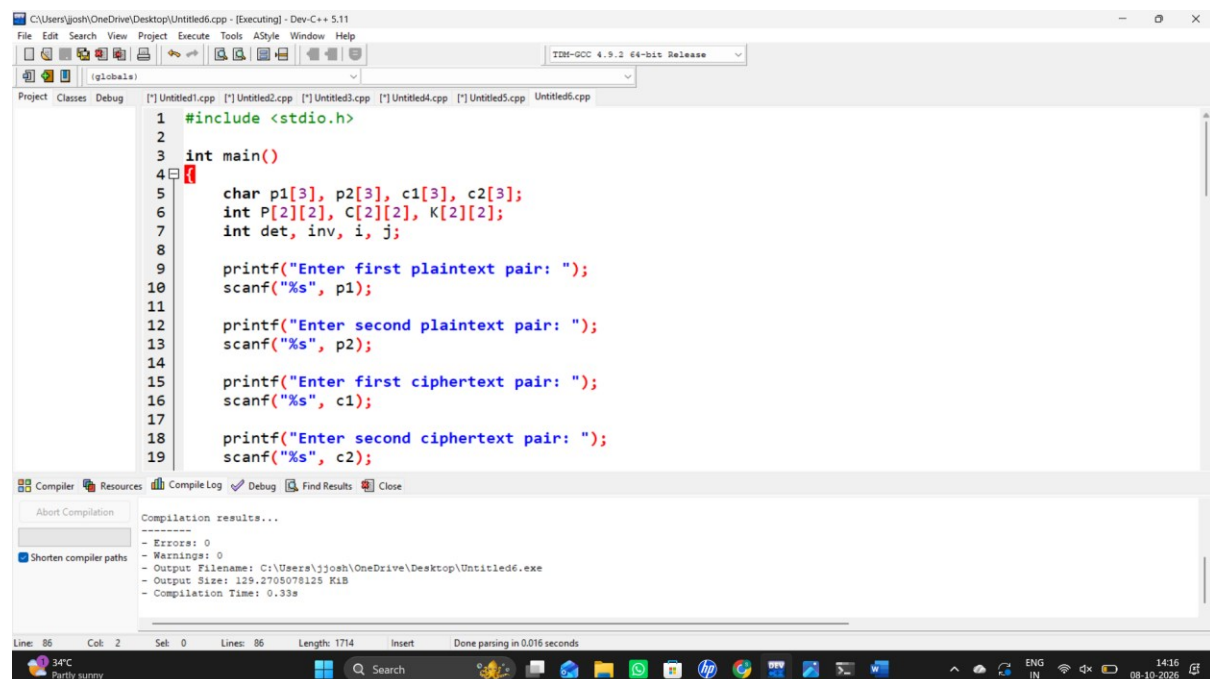
Aim

To demonstrate a known plaintext attack on the Hill cipher using sufficient plaintext-ciphertext pairs.

Algorithm

1. Obtain known plaintext-ciphertext pairs.
2. Convert the letters into numerical values.
3. Arrange the plaintext pairs into a matrix P.
4. Arrange the corresponding ciphertext pairs into a matrix C.
5. Calculate the inverse of P modulo 26.
6. Calculate the key matrix using: $K = CP^{-1} \pmod{26}$
7. Use the obtained key to decrypt other ciphertexts.
8. Display the recovered key.

Program



```
1 #include <stdio.h>
2
3 int main()
4 {
5     char p1[3], p2[3], c1[3], c2[3];
6     int P[2][2], C[2][2], K[2][2];
7     int det, inv, i, j;
8
9     printf("Enter first plaintext pair: ");
10    scanf("%s", p1);
11
12    printf("Enter second plaintext pair: ");
13    scanf("%s", p2);
14
15    printf("Enter first ciphertext pair: ");
16    scanf("%s", c1);
17
18    printf("Enter second ciphertext pair: ");
19    scanf("%s", c2);
```

Compilation results...

- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\josh\OneDrive\Desktop\Untitled6.exe
- Output Size: 129.2705078125 KiB
- Compilation Time: 0.33s

Input/Output

```
C:\Users\josh\OneDrive\Desktop >
Enter first plaintext pair: HI
Enter second plaintext pair: AT
Enter first ciphertext pair: TC
Enter second ciphertext pair: ID

Recovered Key Matrix:
21 18
22 7

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Process exited after 37.38 seconds with return value 0
Press any key to continue . . .
```

Result

Thus, the key matrix of the Hill cipher was successfully recovered using a known plaintext attack.